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B.Tech CSE Core
DBMS Assignment

Section - A

Q1 Fill in the blanks.

(i) The "Entity Relationship Diagram (ER)" Diagram usually represents how entities relate in a Database, offering a clear blueprint of the Data's organizational structure.

(ii) In the "Relational Model", data is organized into tables, with relationship between Entities defined through Foreign key Constraints.

(iii) The "Object-Oriented" Model treats data as a Collection of objects, each with unique Attributes & Methods, Mirroring real word Entities and Interactions.

(iv) "Integrity" Constraints ensures the reliability of data by enforcing rules that must be adhered to maintain the accuracy of Data.

Section-B [Case Study Based]

Q2.

Ans → Data Abstraction in DBMS refers to hiding complex database structure & showing only relevant information to users. In Health care system, it plays an important role in simplifying how Doctors, Nurses & Staff interact with patient records.

→ In a Hospital Environment, large Amount of patients data such as Medical history, Diagnosis, Test results & billing Information are stored in Database of health care professionals had to deal with the complex internal structure of database, it would be difficult and Time Consuming.

→ Data ~~Abstra~~ Abstraction works at Three level:

1) Physical level: A Doctor doesn't need to know how data is stored on the server's hard drive or the specific indexing technique used.

ii) Logical level: A Nurse sees a logical view of "Patient Vitals" which Abstract away the backent table joins between "Patients" and "Medical - history".

iii) View Level: A billing specialist only sees insurance and payment info, while a surgeon sees Medical history and lab results, preventing information overload.

Q3)

Ans) Benefits:

i) Improve Data Security: Sensitive Information of patient is protected because user only Access Data relevant to their roles.

ii) Simplified Data Access: Doctors & Nurses can easily retrieve patient information without understanding Complex Database Structures.

iii) Better Data Management: Data Abstraction allows efficient organisation & Management of large Healthcare datasets.

Challenges:

i) Implementation Complexity: Designing multiple levels of abstraction requires careful planning & Expertise.

ii) System Cost: Developing and Maintaining such systems can be ~~too~~ expensive.

iii) Data Integration Issue: Integration Data from different hospital departments can be challenging.

→ Overall ~~Da~~ despite these challenges, ~~dat~~ Data Abstraction significantly improves the efficiency and security of healthcare database systems.

Section - C [Simulation Based]

Q3) Design a University Database using ER Model.

Teleo Dt.: Pg.:

→ Entities:

i) Student Entity
- → Student_ID [Primary key]
- → Student_Name
- → Mobile No.
- → Department

ii) Course Entity
- → Course_ID [Primary key]
- → Course_Name
- → Credits

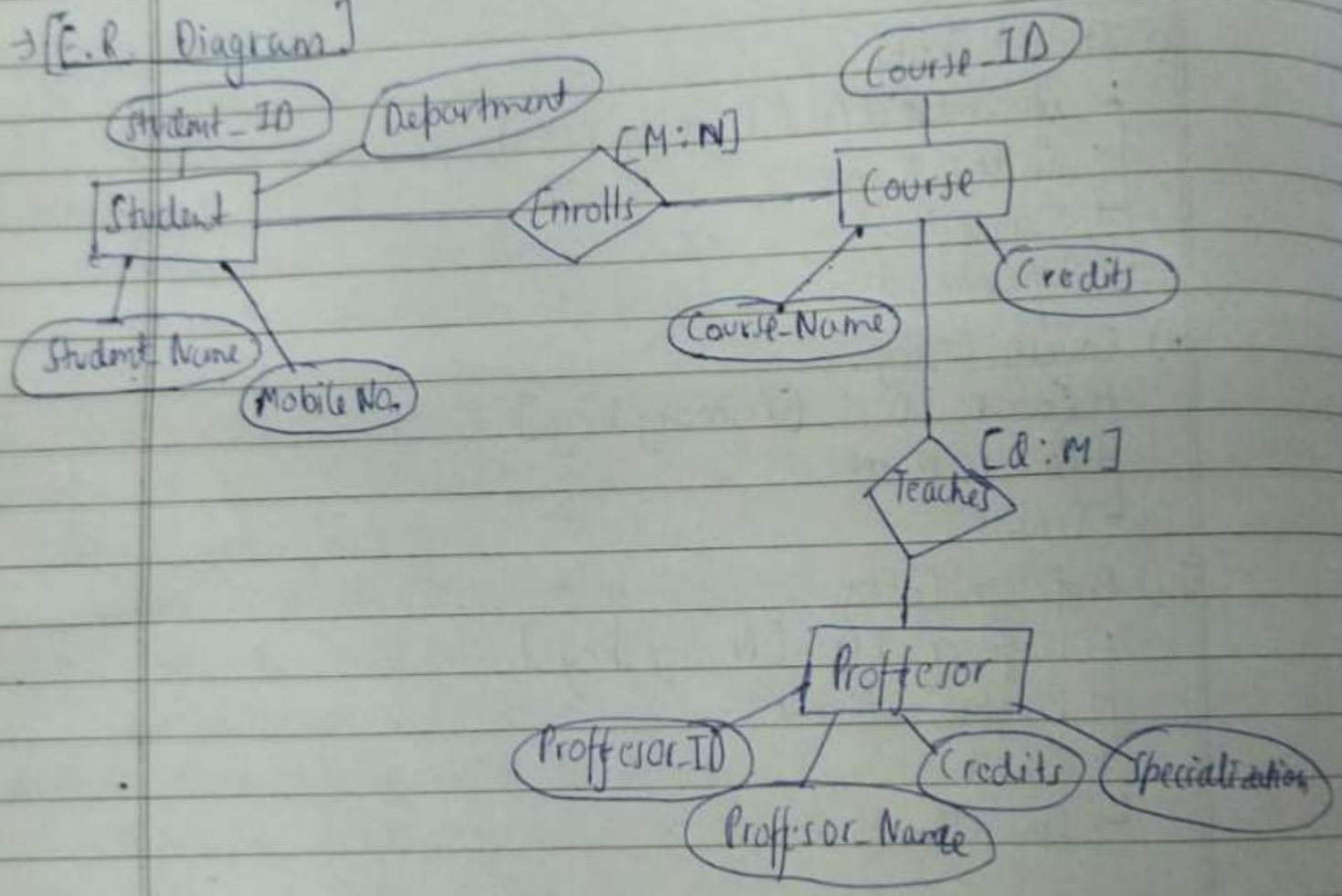
iii) Professor Entity
- → Professor_ID [Primary key]
- → Professor_Name
- → Email
- → Specialization

→ Relationships:

i) Enrollment Relationship [Student + Course]
→ A student can enroll in multiple course & a course can have many students. [Rel. Type → M:N]

ii) Teaching Relationship [Professor → Course]
→ A professor can teach multiple course but a course is taught by one professor. [Rel. Type: One to Many]

→ [E.R. Diagram]



→ This ER Model helps organise University data efficiently, ensuring easy management of student enrollments, course details & proffessor information.